Krishna Kumar Jha

AI/ML Engineer | Generative AI | Machine Learning
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Technical Skills

Programming: Python, Java, C++, SQL
Generative AI / LLM: LLMs, RAG, Prompt Engineering, Embeddings, Semantic Search, Hybrid Retrieval, Vector Databases, LLM Evaluation, Grounding, Hugging Face,LangChain, LangGraph
Machine Learning: Scikit-Learn, PyTorch, TensorFlow, Keras, Machine Learning,Deep Learning, Feature Engineering, Classification, Regression, Model Evaluation
Backend / Data: FastAPI, REST APIs, PostgreSQL, pgvector, MongoDB, Pandas, NumPy
Computer Vision: OpenCV, YOLOv8, Image Processing, Object Detection, Data Augmentation

Projects

AI-Powered Technical Support Resolution Platform **FastAPI | PostgreSQL/pgvector | React**
Generative AI, Hybrid Retrieval, LLM Evaluation

- Built an end-to-end technical support-ticket assistant indexing **217 GitLab documentation pages** into **4,600 chunks**, using Qwen3 embeddings and Nemotron-120B to generate citation-backed technical resolutions.
- Engineered hybrid retrieval with HNSW vector search, PostgreSQL full-text/exact-phrase matching, and Reciprocal Rank Fusion; tuned relevance thresholds on a **51-case benchmark**, achieving **0.87 Recall@5** and **0.88 MRR**.
- Implemented LLM reliability and validation pipelines with JSON-schema enforcement, citation/grounding validation, prompt-injection screening, and failure handling; achieved 93% triage accuracy, 100% schema-valid outputs, with no evaluated unsupported response published across 61 analyses, backed by 517 automated unit, integration, security, and end-to-end tests using Docker Compose.

Seal and Signature Detection System **Python | YOLOv8 | OpenCV | Streamlit**
Computer Vision, Object Detection

- Engineered a document-intelligence system for automated detection of handwritten signatures and official seals using YOLOv8, achieving **97.1% mAP@50** on **4,902 annotated document images**.
- Designed a synthetic document generation pipeline producing **3,000 training samples** to improve model generalization across diverse document layouts; deployed end-to-end inference through Streamlit with **95.4% precision** and **92.6% recall**.

Electricity Demand Forecasting **Python | XGBoost | LSTM | Prophet**
Time-Series Forecasting, Feature Engineering

- Developed hourly electricity-demand forecasting using the PJM dataset with temporal, weather, holiday, and seasonal features; compared Prophet, XGBoost, and LSTM models using MAE, RMSE, and MAPE.
- Built an interactive forecasting dashboard to visualize demand trends, predict future consumption, and recommend generation schedules for resource planning.

Experience

Machine Learning Intern — CodSoft **May 2024 – Jul 2024**
Remote

- Developed and deployed a heart-disease prediction dashboard using Scikit-Learn, Pandas, and Streamlit, supporting real-time inference from patient health metrics.
- Performed data preprocessing, feature scaling, exploratory analysis, and supervised model training; integrated trained models into an interactive end-to-end prediction workflow.

Education

Lovely Professional University, Jalandhar **2022 – 2026**
B.Tech. in Computer Science & Engineering **CGPA: 7.49/10**

H.M. Education Center **2020 – 2022**
Higher Secondary Education **81%**

Certifications

DeepLearning.AI TensorFlow Developer Professional Certificate Coursera, 2026
Oracle Agentic AI Foundations Associate Oracle, 2026